

smid_calculation

May 1, 2026

1 Super Midsize Jet Premium Extraction Audit — Challenger Fleet

Segment: Super Midsize Jet (SMID) | **Fleet:** Challenger series | **Data pipeline:** Strategic Rhombus — `smid_final_wingx_24_25` reference notebook

1.1 Purpose

This notebook identifies the **highest-value pricing windows** across 20 SMID strategic corridors. The analysis mirrors the LJ methodology with three SMID-specific calibrations:

Calibration	LJ value	SMID value	Reason
Velocity floor	15%	12%	Denver-hub corridors distribute demand more evenly across TOD slots
National WUP baseline	4.0%	3.4%	Segment-specific overall WUP SMID long-haul share (ref Cell 5 source notebook)
Stronghold WUP threshold	> 6.5%	> 7.2%	SMID distribution has higher top-end WUP concentration

1.2 Data Sources

File	Records	Coverage
<code>smid_dow_heatmaps.json</code>	140	20 corridors × 7 days-of-week
<code>smid_tod_heatmaps.json</code>	120	20 corridors × 6 time-of-day buckets

Pre-computed corridor aggregates from the Strategic Rhombus pipeline. Raw flight-level data is processed upstream — this notebook ingests summary records only.

1.3 Column Glossary

Column	Definition
tod_market_velocity_pct	% of corridor's annual flights occurring in this TOD window — measures market depth
tod_wup_share_pct	WUP flights ÷ total flights in that window — the core extraction signal
dow_market_velocity_pct	% of corridor's annual flights on this day-of-week
dow_wup_share_pct	WUP flights ÷ total flights on that day-of-week

1.4 Analysis Pipeline

Cell	Output	Purpose
1-3	df_dow_final, df_tod_final	Load and normalize raw heatmap records
4	df_best (20 rows)	Rank-sum selection — one best DOW day + one best TOD slot per corridor
5	df_categorized (20 rows)	Premium extraction classification into 4 tiers
6	df_value (20 rows)	Revenue opportunity quantification per corridor
7-8	df_signals	Active corridors only (No Signal removed)

Loading SMID heatmap data...

DOW records : 140

TOD records : 120

DOW shape : (140, 4)

Corridors : 20

Days : ['Friday', 'Monday', 'Saturday', 'Sunday', 'Thursday', 'Tuesday', 'Wednesday']

	corridor	dimension	dow_market_velocity_pct	\
0	New York City	South Florida	Monday	13.7199
1	New York City	South Florida	Tuesday	12.0561
2	New York City	South Florida	Wednesday	15.3092
3	New York City	South Florida	Thursday	17.3578
4	New York City	South Florida	Friday	16.5632

5	New York City	South Florida	Saturday	10.2930
6	New York City	South Florida	Sunday	14.7008

dow_wup_share_pct	
0	1.6290
1	0.8239
2	1.0543
3	1.0014
4	1.1244
5	1.4475
6	1.4358

TOD shape : (120, 4)

Corridors : 20

Buckets : ['07:00-09:59', '10:00-12:59', '13:00-15:59', '16:00-18:59', '19:00-21:59', 'Off-Peak (Late/Early)']

	corridor		dimension \
0	New York City	South Florida	07:00-09:59
1	New York City	South Florida	10:00-12:59
2	New York City	South Florida	13:00-15:59
3	New York City	South Florida	16:00-18:59
4	New York City	South Florida	19:00-21:59
5	New York City	South Florida	Off-Peak (Late/Early)

	tod_market_velocity_pct	tod_wup_share_pct
0	18.1276	0.7534
1	28.4579	1.3089
2	24.1122	1.3388
3	18.3760	1.6216
4	7.5490	0.4934
5	3.3772	1.1029

1.5 Data Structure After Normalization

Each JSON file contains one record per corridor-dimension pair. Example TOD record:

```
{ "corridor": "South Florida  Denver",
  "dimension": "10:00-12:59",
  "tod_market_velocity_pct": 42.54,
  "tod_wup_share_pct": 9.23 }
```

SMID-specific data characteristic — Denver hub concentration:

The SMID corridor set is Denver-hub heavy, which creates a distinct demand pattern: - Denver-hub corridors often have **flatter velocity distributions** across TOD slots (no single dominant peak) - This means the rank-sum “winner” may have a velocity of 14–15%, below the LJ floor of 15% - A 14.6% velocity on Denver → Boston is still a genuine active window — the 12% floor is calibrated accordingly - Denver → Boston (vel=14.6%, wup=8.61%) is correctly classified as **Stronghold**: WUP concentration is exceptional ($8.61\% = 2.53\times$ national average) even if velocity is moderate

1.6 Rank-Sum Dimension Selection — How It Works

For each corridor we pick **one DOW day** and **one TOD slot** where velocity and WUP share are simultaneously highest. A simple “pick the max” approach fails because the fastest slot is rarely the one with the highest share. Instead we use a **rank-sum**: rank each slot independently on both metrics, add the ranks, and pick the slot with the lowest total.

1.6.1 Algorithm (applied separately to DOW and TOD)

1. For every slot in a corridor, assign **vel_rank** — rank 1 = **fastest** (descending).
2. For every slot in a corridor, assign **wup_rank** — rank 1 = **highest WUP share** (descending).
3. **combined_rank** = **vel_rank** + **wup_rank**
4. Pick the slot with the **lowest combined_rank**.
5. Tiebreaker (equal **combined_rank**): highest **wup_share**, then highest **velocity**.

A slot that ranks #1 on both gets **combined_rank** = 2 — the theoretical minimum.

1.6.2 Hand-Trace Example — South Florida → Denver (SMID’s top corridor)

Time Slot	tod_velocity	tod_wup	vel_rank	wup_rank	combined_rank
07:00–09:59	14.2%	3.1%	5	3	8
10:00–12:59	42.5%	9.2%	1	1	2 ← winner
13:00–15:59	21.8%	5.7%	3	2	5
16:00–18:59	31.4%	2.4%	2	4	6
19:00–21:59	12.1%	1.8%	4	5	9
Off-Peak	3.1%	0.6%	6	6	12

10:00–12:59 ranks **#1 on velocity AND #1 on WUP share** → **combined_rank** = 2 → selected. This is the ideal outcome: both signals peak simultaneously.

1.6.3 What the rank-sum protects against

Scenario	Pure-velocity pick	Pure-WUP pick	Rank-sum pick
Fastest slot has near-zero WUP share	wrong slot	—	balanced
Best-WUP slot is off-peak (vel 2%)	—	wrong slot	balanced
Both metrics peak at the same slot	correct	correct	correct
Metrics peak at different slots	misses WUP	misses demand	finds best balance

1.6.4 SMID Edge Case — Denver → Boston (vel = 14.6%, wup = 8.61%)

Denver → Boston wins at the **16:00–18:59 slot** with velocity 14.6% — below the LJ floor of 15%. This is intentional: the 12% floor was set specifically for Denver-hub corridors. The WUP concentration of 8.61% (SLI = 2.53) is the second-strongest in the SMID set. The 16:00–18:59 window is the best *balanced* slot — other slots have higher velocity but significantly lower WUP share. Rank-sum correctly identifies this as the actionable window.

1.6.5 SMID-Specific Parameters

Parameter	Value	Source
Velocity floor	12%	Denver-hub demand distributes more evenly across slots
National WUP baseline	3.4%	$\text{overall_wup_share} = (\text{total_wup_missions} / \text{total_long_haul}) * 100$ — ref notebook Cell 5
SLI formula	$\text{tod_wup_share_pct} / 3.4$	Measures WUP's concentration vs SMID segment average

Best-dimension table: (20, 7) (expect 20 x 7)

		corridor	dow_dimension	tod_dimension \
0	Atlanta	Denver	Thursday	13:00-15:59
1	Boston	Denver	Sunday	07:00-09:59
2	Boston	South Florida	Thursday	10:00-12:59
3	Chicago	Denver	Thursday	16:00-18:59
4	Chicago	South Florida	Thursday	10:00-12:59
5		DMV Denver	Sunday	13:00-15:59
6	Dallas	New York City	Wednesday	07:00-09:59
7		Denver Boston	Monday	16:00-18:59
8	Denver	New York City	Sunday	10:00-12:59
9	Denver	South Florida	Saturday	10:00-12:59
10	Minneapolis	South Florida	Friday	07:00-09:59
11		New York City Dallas	Tuesday	07:00-09:59
12		New York City Denver	Friday	10:00-12:59
13	New York City	South Florida	Monday	16:00-18:59
14		South Florida Boston	Friday	10:00-12:59
15		South Florida Chicago	Thursday	13:00-15:59
16		South Florida Dallas	Sunday	07:00-09:59
17		South Florida Denver	Friday	10:00-12:59
18	South Florida	Minneapolis	Wednesday	10:00-12:59
19	South Florida	New York City	Sunday	10:00-12:59

	dow_velocity_pct	tod_velocity_pct	dow_wup_share_pct	tod_wup_share_pct
0	16.5734	28.0420	7.5949	7.9800
1	17.6390	23.8161	9.3385	7.7810
2	15.6514	32.2868	3.4431	4.6444
3	16.3307	16.4858	1.8987	3.7618
4	16.2087	33.6348	2.3605	1.8614
5	12.8511	24.5787	9.2896	6.2857
6	15.8099	17.8952	3.0675	2.1680
7	15.6162	14.6359	10.7623	8.6124
8	19.5061	38.6880	7.1611	6.7054
9	16.0600	41.3888	13.1429	8.3518
10	15.8773	20.9473	2.9412	2.5478
11	15.7274	12.4509	2.7778	4.2105
12	16.7411	38.1200	5.6296	6.6363
13	13.7199	18.3760	1.6290	1.6216
14	15.5011	33.7555	2.8419	3.0179
15	13.6752	27.4454	2.2569	2.3356
16	15.9703	17.5487	1.1628	1.8519
17	14.7040	42.5354	10.5033	9.2284
18	14.4909	33.5415	4.3103	3.9106
19	19.8702	27.9726	0.8826	0.6270

1.7 Step 3 — Premium Extraction Classification (SMID-Calibrated)

1.7.1 Threshold derivation

All 20 SMID corridor TOD winners were sorted by `tod_wup_share_pct` descending. Cut points placed at **natural distributional gaps**:

Rank	Corridor	WUP%	Gap to next
5	Boston → Denver	7.78%	↓ 1.07 pp gap
6	Denver → New York City	6.71%	
8	DMV → Denver	6.29%	↓ 1.65 pp gap
9	Boston → South Florida	4.64%	
13	South Florida → Boston	3.02%	↓ 0.47 pp gap
14	Minneapolis → South Florida	2.55%	

Cut points: > **7.2%** (top tier), **4.0%** (second tier), **2.25%** (third tier).

1.7.2 Category definitions (SMID)

Category	Condition	Revenue action
Stronghold Extraction	wup > 7.2% AND velocity > 12%	Maximum premium +25–30% — dominant Denver-hub WUP position
Surge Capture	4.0% wup 7.2% AND velocity > 12%	Aggressive push +15–20% — strong and active windows

Category	Condition	Revenue action
Niche Protection	2.25% wup < 4.0%	Moderate uplift +10–15% — defend share, no velocity gate needed
Velocity Opportunity	velocity > 25% AND 2.0% wup < 2.25%	Selective push — high demand, early-stage WUP presence
No Signal	all other	No pricing action

1.7.3 Dual-criterion gate (same as LJ)

Every active SMID candidate satisfies **both**: - Strong market demand: velocity 12% (SMID-calibrated floor) - Meaningful WUP presence: wup 2.0% (no sub-2% extractions)

Note: Velocity Opportunity has **0 qualifiers** in the SMID dataset — both candidates (Chicago → SF at 1.86%, SF → New York City at 0.63%) fall below the 2.0% WUP floor and are correctly assigned to No Signal.

SMID PREMIUM EXTRACTION CATEGORY DISTRIBUTION

category

Stronghold Extraction	5
Surge Capture	5
Niche Protection	5
Velocity Opportunity	0
No Signal	5

		corridor	dow_dimension	dow_velocity_pct	\
0	South Florida	Denver	Friday	14.7040	
1	Denver	South Florida	Saturday	16.0600	
2	Atlanta	Denver	Thursday	16.5734	
3	Boston	Denver	Sunday	17.6390	
4	Denver	Boston	Monday	15.6162	
5	Denver	New York City	Sunday	19.5061	
6	New York City	Denver	Friday	16.7411	
7	Boston	South Florida	Thursday	15.6514	
8	DMV	Denver	Sunday	12.8511	
9	New York City	Dallas	Tuesday	15.7274	
10	South Florida	Boston	Friday	15.5011	
11	South Florida	Minneapolis	Wednesday	14.4909	
12	South Florida	Chicago	Thursday	13.6752	
13	Minneapolis	South Florida	Friday	15.8773	
14	Chicago	Denver	Thursday	16.3307	
15	Chicago	South Florida	Thursday	16.2087	
16	South Florida	New York City	Sunday	19.8702	
17	New York City	South Florida	Monday	13.7199	
18	Dallas	New York City	Wednesday	15.8099	

19 South Florida Dallas Sunday 15.9703

	dow_wup_share_pct	tod_dimension	tod_velocity_pct	tod_wup_share_pct	\
0	10.5033	10:00-12:59	42.5354	9.2284	
1	13.1429	10:00-12:59	41.3888	8.3518	
2	7.5949	13:00-15:59	28.0420	7.9800	
3	9.3385	07:00-09:59	23.8161	7.7810	
4	10.7623	16:00-18:59	14.6359	8.6124	
5	7.1611	10:00-12:59	38.6880	6.7054	
6	5.6296	10:00-12:59	38.1200	6.6363	
7	3.4431	10:00-12:59	32.2868	4.6444	
8	9.2896	13:00-15:59	24.5787	6.2857	
9	2.7778	07:00-09:59	12.4509	4.2105	
10	2.8419	10:00-12:59	33.7555	3.0179	
11	4.3103	10:00-12:59	33.5415	3.9106	
12	2.2569	13:00-15:59	27.4454	2.3356	
13	2.9412	07:00-09:59	20.9473	2.5478	
14	1.8987	16:00-18:59	16.4858	3.7618	
15	2.3605	10:00-12:59	33.6348	1.8614	
16	0.8826	10:00-12:59	27.9726	0.6270	
17	1.6290	16:00-18:59	18.3760	1.6216	
18	3.0675	07:00-09:59	17.8952	2.1680	
19	1.1628	07:00-09:59	17.5487	1.8519	

	category
0	Stronghold Extraction
1	Stronghold Extraction
2	Stronghold Extraction
3	Stronghold Extraction
4	Stronghold Extraction
5	Surge Capture
6	Surge Capture
7	Surge Capture
8	Surge Capture
9	Surge Capture
10	Niche Protection
11	Niche Protection
12	Niche Protection
13	Niche Protection
14	Niche Protection
15	No Signal
16	No Signal
17	No Signal
18	No Signal
19	No Signal

1.8 Step 4 — Revenue Opportunity Quantification (SMID / Challenger)

1.8.1 Fleet parameters (Q1 Challenger Fleet)

Parameter	Low Scenario (10T)	High Scenario (17T)
Fleet hours (Q1)	2,600 hrs	4,420 hrs
Active tails	10	17
Hours per tail (Q1)	260 hrs	260 hrs
Hourly rate	\$11,000	\$11,000
Corridor allocation	20 corridors	20 corridors

1.8.2 SLI — Surge Loyalty Index (SMID)

$SLI = \text{tod_wup_share_pct} / \text{NATIONAL_BASELINE}$ where $\text{NATIONAL_BASELINE} = 3.4\%$ (overall WUP SMID long-haul share, verified from reference notebook Cell 5: $\text{overall_wup_share} = (\text{total_wup_missions} / \text{total_long_haul}) * 100$)

SLI value	Interpretation
< 1.0	WUP underweight vs SMID market average
1.0	WUP exactly at SMID market average
1.5	WUP 50% above market average — moderate signal
2.5+	WUP more than 2.5× market average — Stronghold threshold

Example: South Florida → Denver at 10:00–12:59 has $\text{wup} = 9.23\% \rightarrow SLI = 9.23 / 3.4 = \mathbf{2.71}$ (Stronghold, +30% toggle)

1.8.3 Hours-in-Window Formula

$$\text{hours} = (\text{Fleet_hrs} / \text{N_corridors}) \times (\text{tod_velocity_pct} / 100) \times SLI$$

Factor	Low	High	Meaning
Fleet_hrs / N_corridors	130 hrs	221 hrs	Per-corridor fleet allocation
× tod_velocity_pct / 100	varies	varies	Fraction in the peak TOD window
× SLI	varies	varies	WUP concentration multiplier

1.8.4 Variable Toggle — Pricing Uplift within Category

Category	Condition for high toggle	High	Low
Stronghold Extraction	$SLI > 2.5$	+30%	+25%

Category	Condition for high toggle	High	Low
Surge Capture	—	+20%	+20%
Niche Protection	wup > 3.3%	+15%	+10%
Velocity Opportunity	velocity > 33%	+20%	+15%

1.8.5 Value Formula

$$\text{value} = \text{hours_in_window} \times \text{hourly_rate} \times \text{toggle_pct}$$

Incremental revenue from applying the pricing uplift during the peak extraction window. The SLI multiplier ensures opportunity sizing is proportional to WUP's actual presence — high-SLI corridors capture more revenue-eligible hours from the same fleet allocation.

VALUE BY CATEGORY (Q1 Challenger Fleet)

	value_low	value_high
category		
Stronghold Extraction	\$1,448,150	\$2,460,975
Surge Capture	\$731,280	\$1,243,220
Niche Protection	\$214,445	\$364,540
No Signal	\$0	\$0

	corridor		category	tod_dimension \
0	South Florida	Denver	Stronghold Extraction	10:00-12:59
1	Denver	South Florida	Stronghold Extraction	10:00-12:59
2	Atlanta	Denver	Stronghold Extraction	13:00-15:59
3	Boston	Denver	Stronghold Extraction	07:00-09:59
4	Denver	Boston	Stronghold Extraction	16:00-18:59
5	Denver	New York City	Surge Capture	10:00-12:59
6	New York City	Denver	Surge Capture	10:00-12:59
7	Boston	South Florida	Surge Capture	10:00-12:59
8	DMV	Denver	Surge Capture	13:00-15:59
9	New York City	Dallas	Surge Capture	07:00-09:59
10	South Florida	Boston	Niche Protection	10:00-12:59
11	South Florida	Minneapolis	Niche Protection	10:00-12:59
12	South Florida	Chicago	Niche Protection	13:00-15:59
13	Minneapolis	South Florida	Niche Protection	07:00-09:59
14	Chicago	Denver	Niche Protection	16:00-18:59
15	Chicago	South Florida	No Signal	10:00-12:59
16	South Florida	New York City	No Signal	10:00-12:59
17	New York City	South Florida	No Signal	16:00-18:59
18	Dallas	New York City	No Signal	07:00-09:59
19	South Florida	Dallas	No Signal	07:00-09:59

	tod_velocity_pct	tod_wup_share_pct	sli	toggle_pct	hours_low	\
0	42.5354	9.2284	2.71	0.30	149.9	
1	41.3888	8.3518	2.46	0.25	132.4	
2	28.0420	7.9800	2.35	0.25	85.7	
3	23.8161	7.7810	2.29	0.25	70.9	
4	14.6359	8.6124	2.53	0.30	48.1	
5	38.6880	6.7054	1.97	0.20	99.1	
6	38.1200	6.6363	1.95	0.20	96.6	
7	32.2868	4.6444	1.37	0.20	57.5	
8	24.5787	6.2857	1.85	0.20	59.1	
9	12.4509	4.2105	1.24	0.20	20.1	
10	33.7555	3.0179	0.89	0.10	39.1	
11	33.5415	3.9106	1.15	0.15	50.1	
12	27.4454	2.3356	0.69	0.10	24.6	
13	20.9473	2.5478	0.75	0.10	20.4	
14	16.4858	3.7618	1.11	0.15	23.8	
15	33.6348	1.8614	0.55	0.00	24.0	
16	27.9726	0.6270	0.18	0.00	6.5	
17	18.3760	1.6216	0.48	0.00	11.5	
18	17.8952	2.1680	0.64	0.00	14.9	
19	17.5487	1.8519	0.54	0.00	12.3	

	hours_high	value_low	value_high
0	254.7	\$494,670	\$840,510
1	225.0	\$364,100	\$618,750
2	145.6	\$235,675	\$400,400
3	120.5	\$194,975	\$331,375
4	81.8	\$158,730	\$269,940
5	168.4	\$218,020	\$370,480
6	164.3	\$212,520	\$361,460
7	97.8	\$126,500	\$215,160
8	100.5	\$130,020	\$221,100
9	34.1	\$44,220	\$75,020
10	66.4	\$43,010	\$73,040
11	85.2	\$82,665	\$140,580
12	41.9	\$27,060	\$46,090
13	34.7	\$22,440	\$38,170
14	40.4	\$39,270	\$66,660
15	40.9	\$0	\$0
16	11.1	\$0	\$0
17	19.5	\$0	\$0
18	25.3	\$0	\$0
19	20.9	\$0	\$0

Active corridors : 15 / 20

	corridor		category	tod_dimension	\
0	South Florida	Denver	Stronghold Extraction	10:00-12:59	
1	Denver	South Florida	Stronghold Extraction	10:00-12:59	

2	Atlanta	Denver	Stronghold Extraction	13:00-15:59
3	Boston	Denver	Stronghold Extraction	07:00-09:59
4	Denver	Boston	Stronghold Extraction	16:00-18:59
5	Denver	New York City	Surge Capture	10:00-12:59
6	New York City	Denver	Surge Capture	10:00-12:59
7	Boston	South Florida	Surge Capture	10:00-12:59
8	DMV	Denver	Surge Capture	13:00-15:59
9	New York City	Dallas	Surge Capture	07:00-09:59
10	South Florida	Boston	Niche Protection	10:00-12:59
11	South Florida	Minneapolis	Niche Protection	10:00-12:59
12	South Florida	Chicago	Niche Protection	13:00-15:59
13	Minneapolis	South Florida	Niche Protection	07:00-09:59
14	Chicago	Denver	Niche Protection	16:00-18:59

	tod_velocity_pct	tod_wup_share_pct	sli	toggle_pct	hours_low \
0	42.5354	9.2284	2.71	0.30	149.9
1	41.3888	8.3518	2.46	0.25	132.4
2	28.0420	7.9800	2.35	0.25	85.7
3	23.8161	7.7810	2.29	0.25	70.9
4	14.6359	8.6124	2.53	0.30	48.1
5	38.6880	6.7054	1.97	0.20	99.1
6	38.1200	6.6363	1.95	0.20	96.6
7	32.2868	4.6444	1.37	0.20	57.5
8	24.5787	6.2857	1.85	0.20	59.1
9	12.4509	4.2105	1.24	0.20	20.1
10	33.7555	3.0179	0.89	0.10	39.1
11	33.5415	3.9106	1.15	0.15	50.1
12	27.4454	2.3356	0.69	0.10	24.6
13	20.9473	2.5478	0.75	0.10	20.4
14	16.4858	3.7618	1.11	0.15	23.8

	hours_high	value_low	value_high
0	254.7	\$494,670	\$840,510
1	225.0	\$364,100	\$618,750
2	145.6	\$235,675	\$400,400
3	120.5	\$194,975	\$331,375
4	81.8	\$158,730	\$269,940
5	168.4	\$218,020	\$370,480
6	164.3	\$212,520	\$361,460
7	97.8	\$126,500	\$215,160
8	100.5	\$130,020	\$221,100
9	34.1	\$44,220	\$75,020
10	66.4	\$43,010	\$73,040
11	85.2	\$82,665	\$140,580
12	41.9	\$27,060	\$46,090
13	34.7	\$22,440	\$38,170
14	40.4	\$39,270	\$66,660

<IPython.core.display.HTML object>

1.9 Results Interpretation Guide

1.9.1 Priority order for revenue management action

Priority	Category	Corridors	Action
1	Stronghold Extraction	5	Maximum toggles immediately — Denver-hub WUP dominance is clear
2	Surge Capture	5	Deploy within the week — strong WUP + active demand
3	Niche Protection	5	Schedule for next cycle — moderate signal, protect before competition moves
4	No Signal	5	Monitor only — insufficient dual-criterion evidence

1.9.2 Key SMID finding

The SMID set is heavily **Denver-hub concentrated** — 8 of the 10 Stronghold + Surge corridors involve Denver as an origin or destination. This reflects WUP’s outsized concentration in Denver-routed SMID traffic. Revenue management should treat Denver-hub windows as a **franchise asset** to defend.

1.9.3 Low vs High scenario

Low = 10 tails × 260 hrs = 2,600 Q1 hours. High = 17 tails × 260 hrs = 4,420 Q1 hours. Value scales linearly — the toggle recommendations and corridor priorities are identical in both scenarios.